Two-Task Competition in Low-Rank RNNs

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Full Experiment Report
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Parameters:
  N = 500   |   n_seeds = 5   |   t_max = 100.0
  dt = 0.05  |   seed = 42          |   D_mean = 2.5

Contents:
  Families  — Multi-seed deep-dives for all three blueprint families
    1. Frequency (ω):  S vs Δω, sensitivity dS/dΔω vs D_mean
    2. Amplitude (γ):  S vs Δγ, sensitivity dS/dΔγ vs D_mean
    3. Shape (ε):      S vs Δε, sensitivity dS/dΔε vs D_mean

  Comparison — Cross-regime experiments
    LC freq, FP rotation, FP coupling, cross-comparison
    Seed-averaged heatmaps for Δω and Δb
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Families 1–3: Multi-Seed Deep-Dives

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For each of the three blueprint families:

  Family 1 — Frequency:   A(ω) = 0.8 [[1, ω], [-ω, 1]],       base ω = 0.5
  Family 2 — Amplitude:   A(γ) = γ   [[1, 0.5], [-0.5, 1]],    base γ = 0.8
  Family 3 — Shape:       A(ε) = 0.8 [[ε, 0.5], [-0.5, 1/ε]],  base ε = 1.0

Two analyses per family (multi-seed averaged):
  1. S vs Δ(feature) at several fixed ΔD values, with ±1σ bands
     Right panel: ΔD = 0 only — does feature difference alone
     drive winner-take-all?
  2. Sensitivity dS/d(Δfeature) at Δfeature = 0 as a function
     of mean connection strength D_mean
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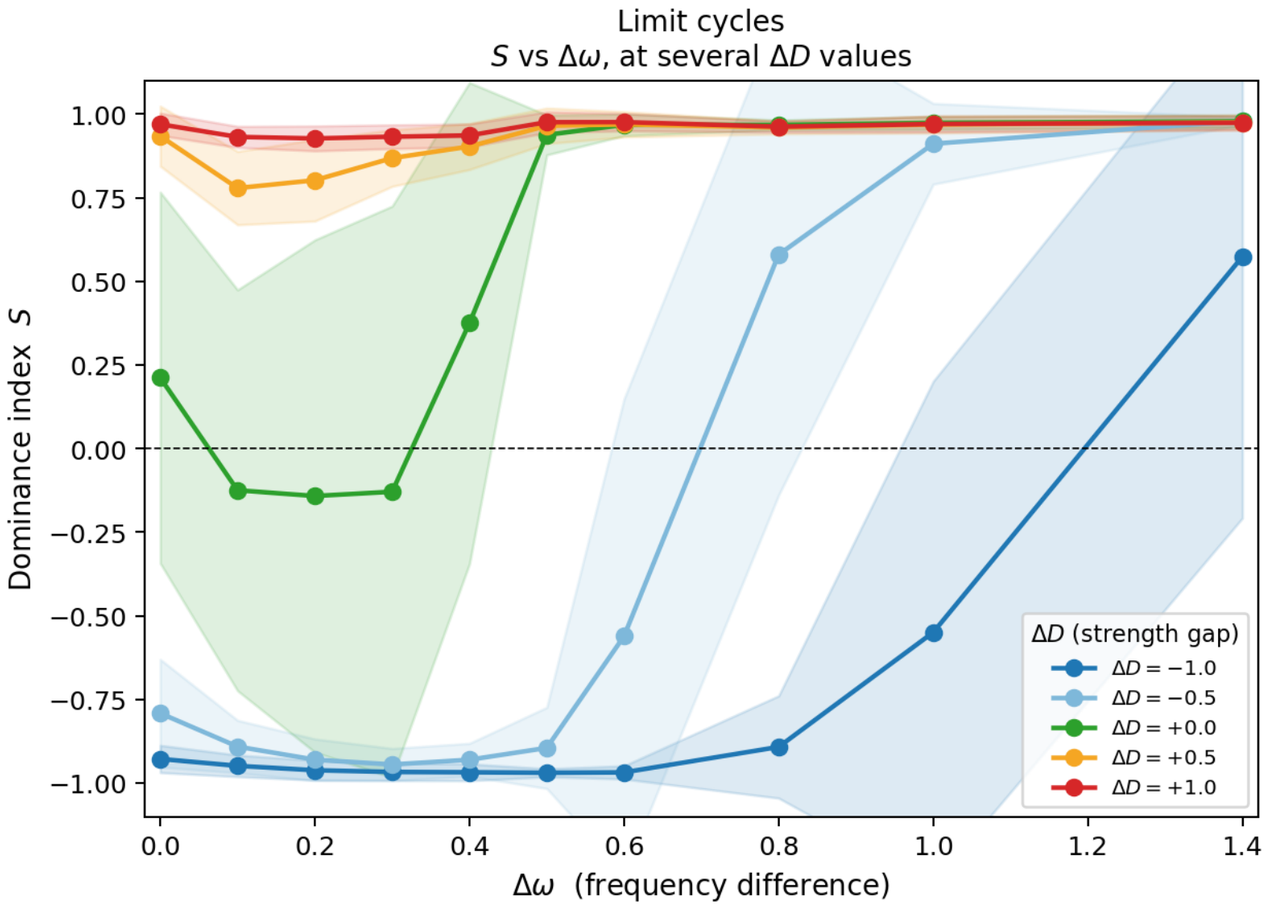
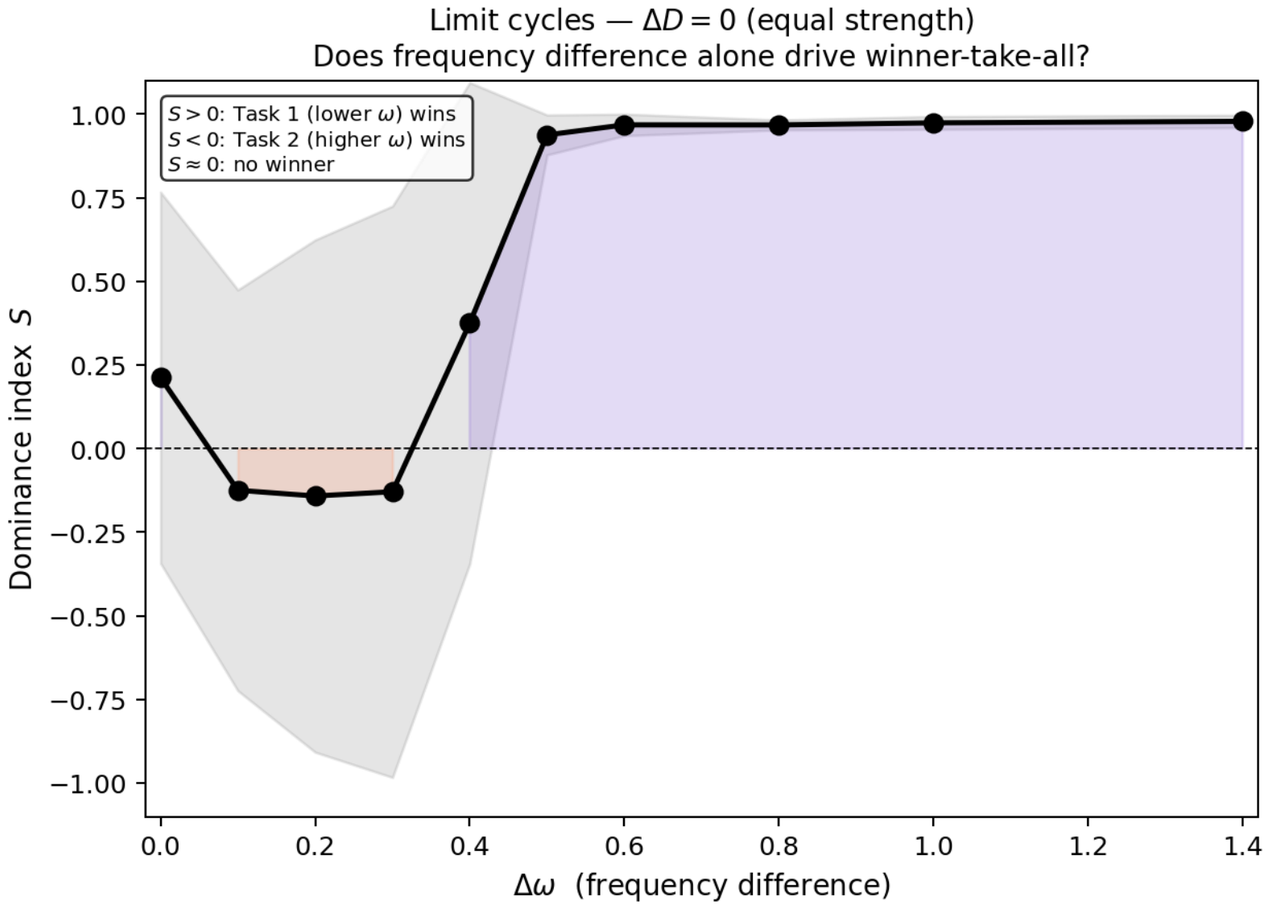
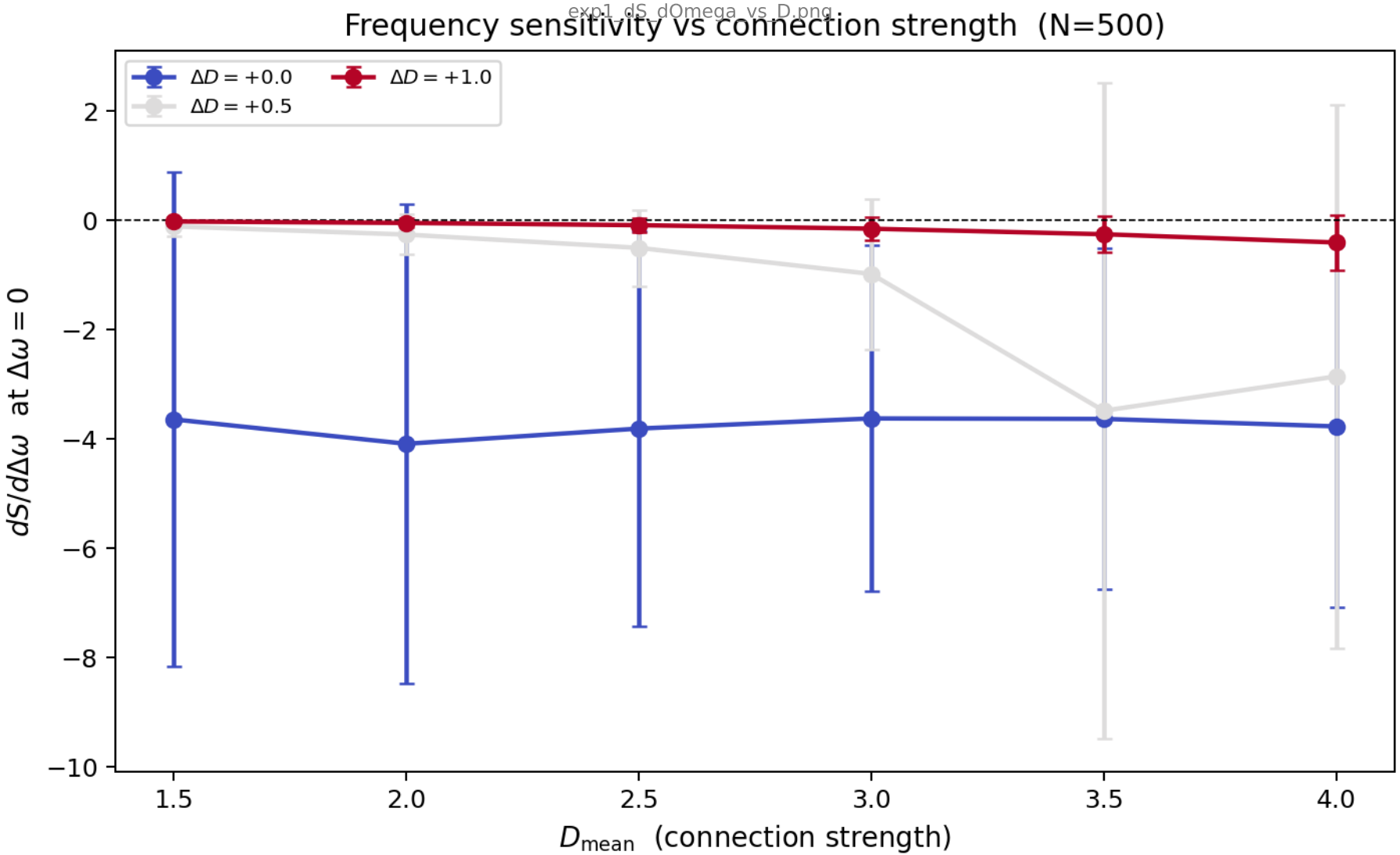
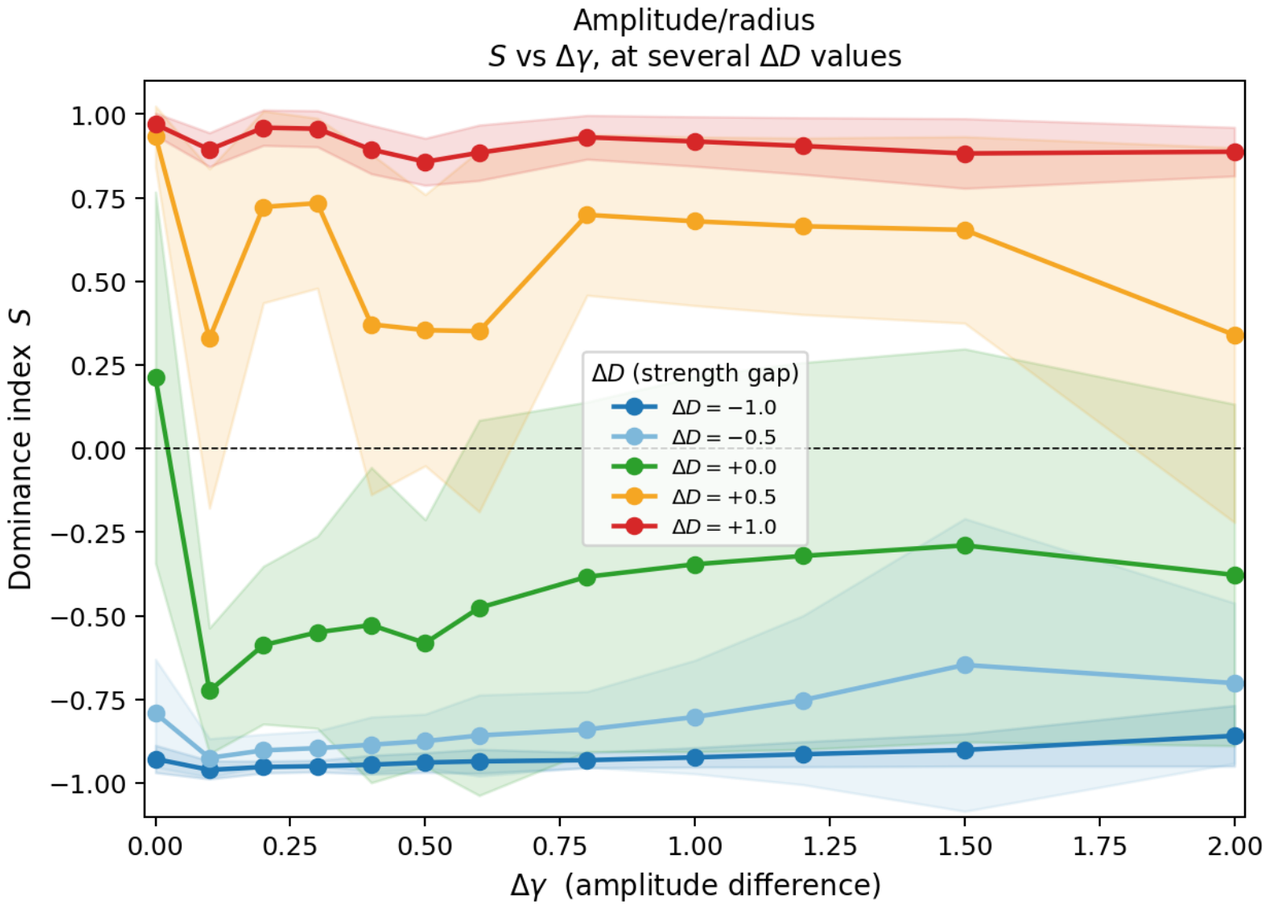
Exp 1 — Effect of frequency difference on winner-take-all (N=500)

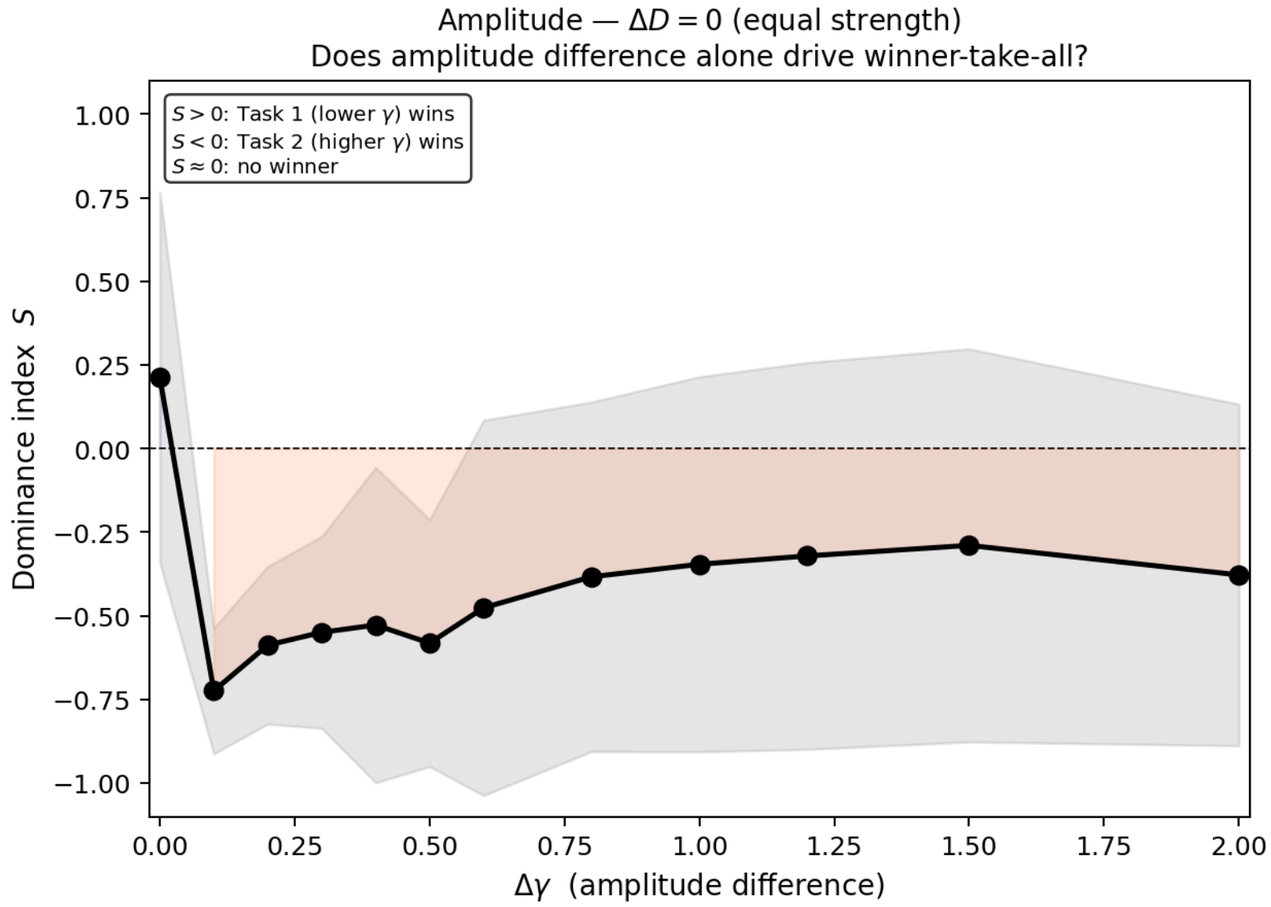


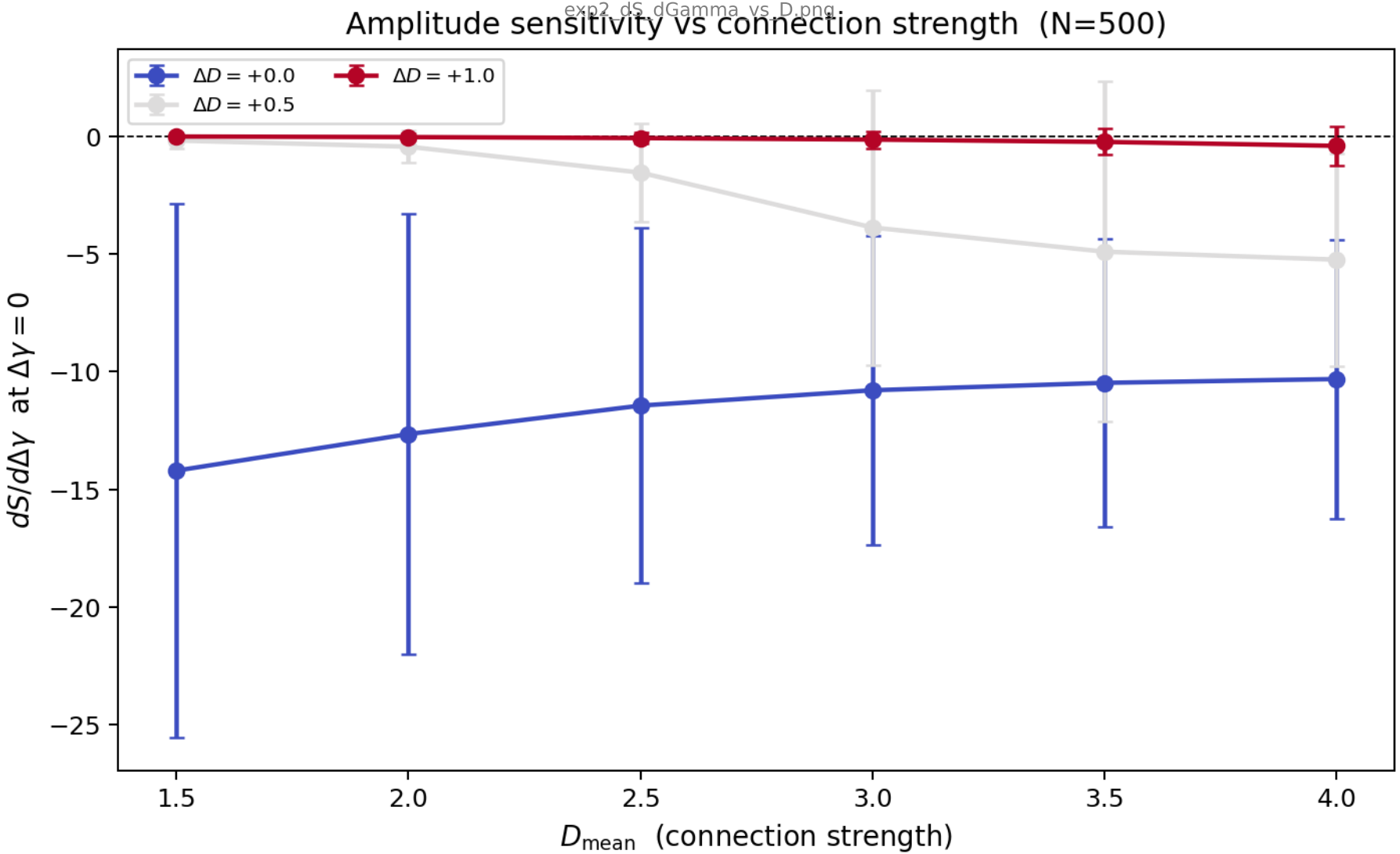



exp1_dS_dOmega_vs_D.png
Frequency sensitivity vs connection strength (N=500)
$\Delta D = +0.0$
$\Delta D = +0.5$
$\Delta D = +1.0$
$dS/d\Delta\omega$ at $\Delta\omega = 0$
D_{mean} (connection strength)
2
0
−2
−4
−6
−8
−10
1.5
2.0
2.5
3.0
3.5
4.0

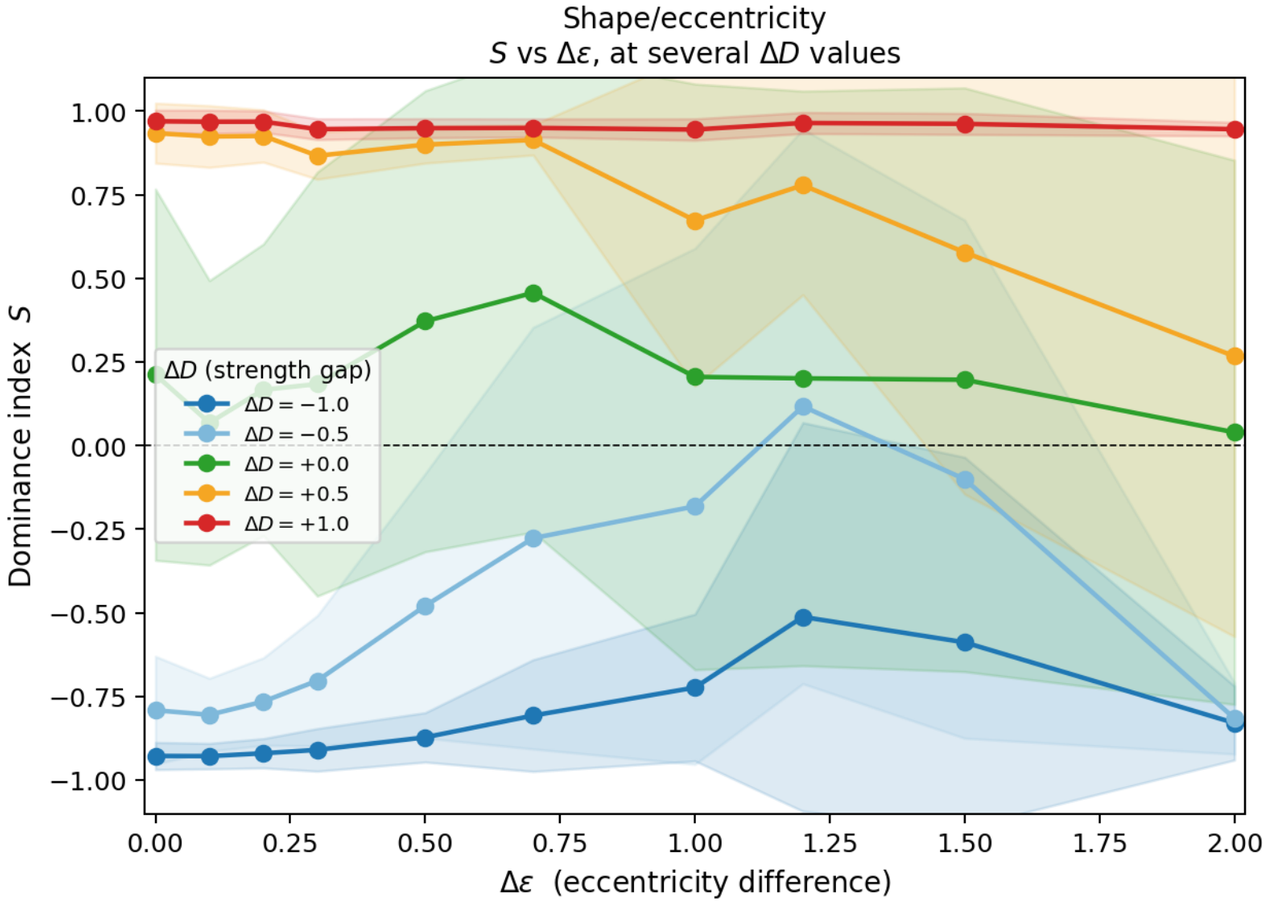
Exp 2 — Effect of amplitude difference on winner-take-all (N=500)

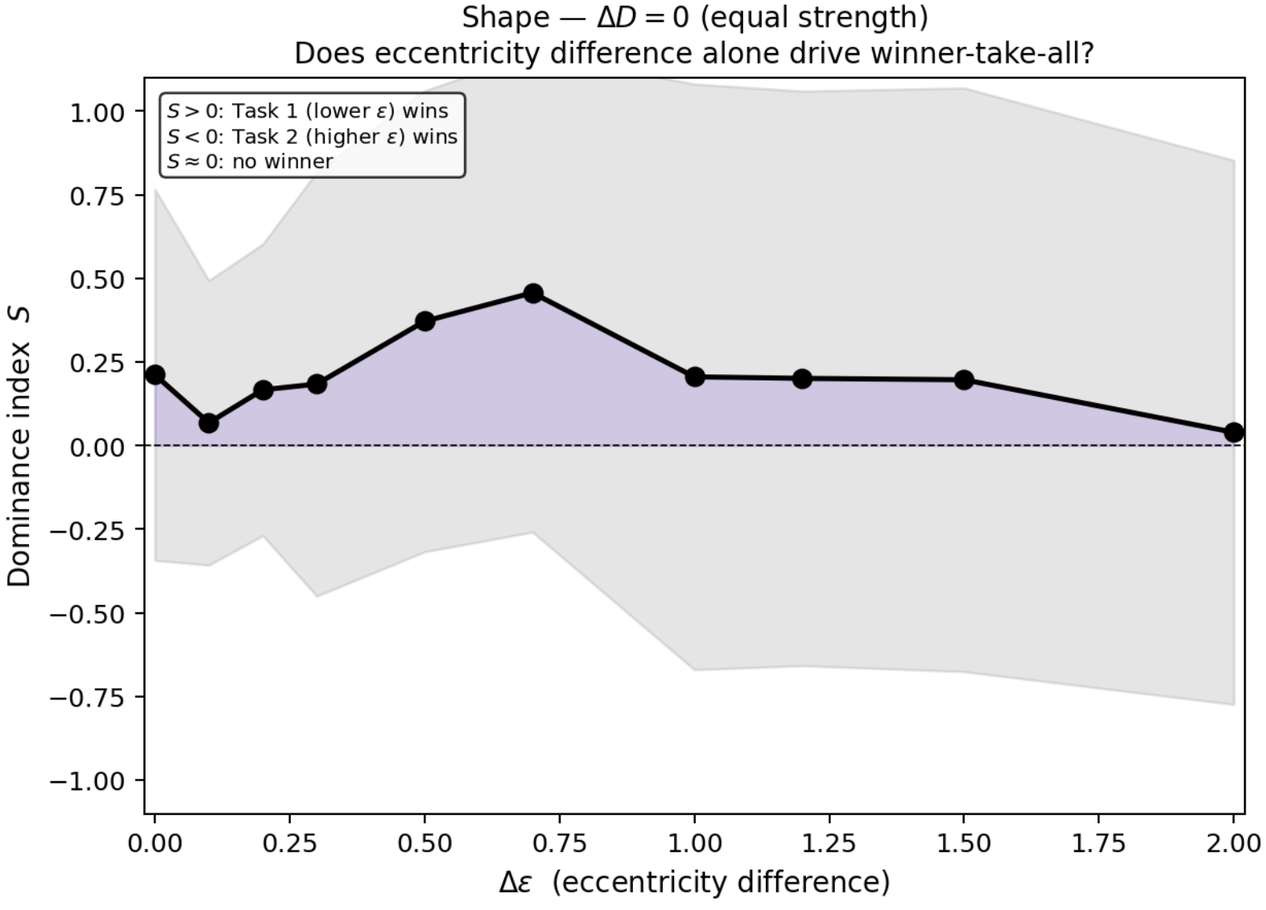




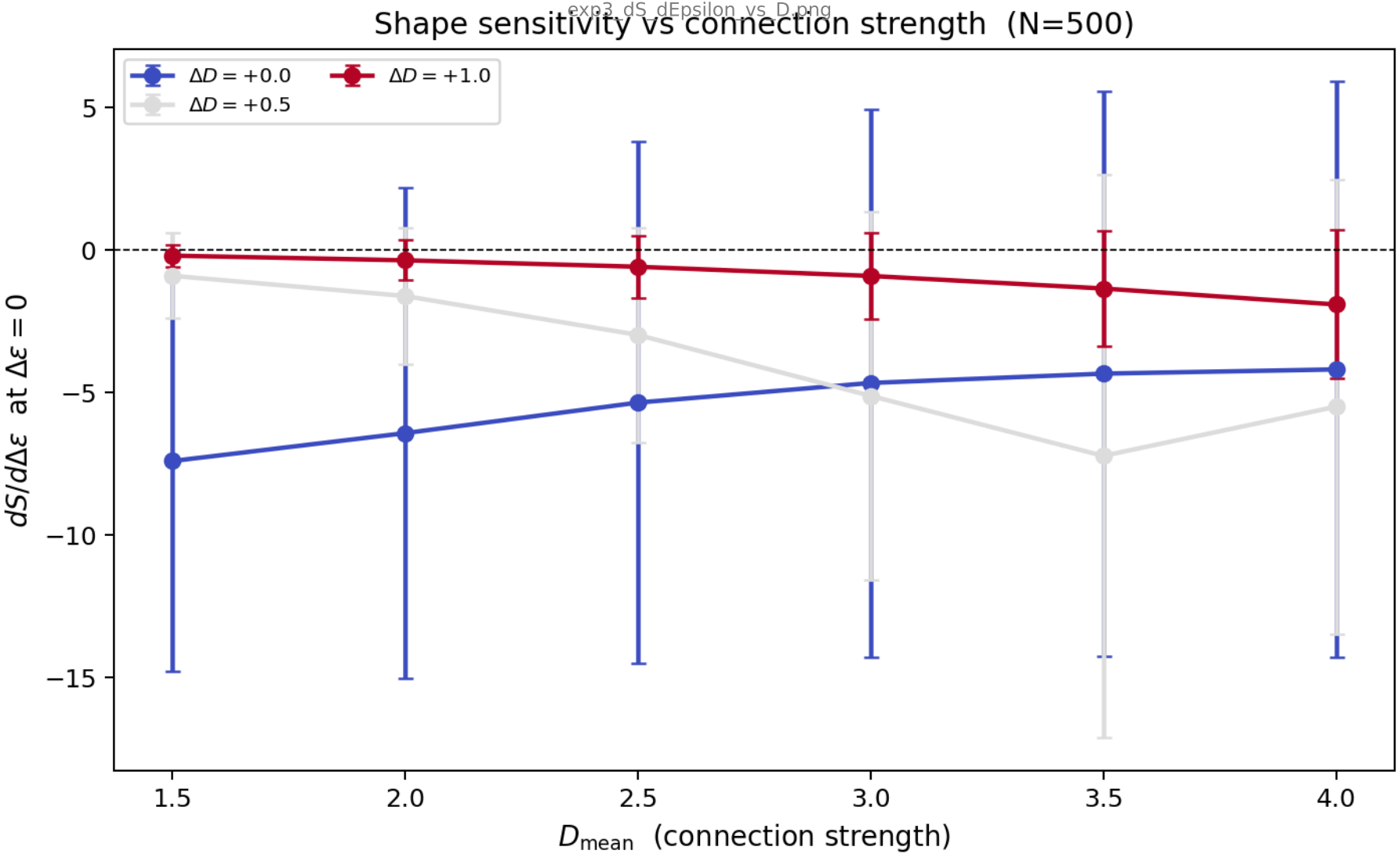

exp2_dS_dGamma_vs_D.png
Amplitude sensitivity vs connection strength (N=500)
$\Delta D = +0.0$
$\Delta D = +0.5$
$\Delta D = +1.0$
0
−5
−10
−15
−20
−25
$dS/d\Delta\gamma$ at $\Delta\gamma = 0$
1.5
2.0
2.5
3.0
3.5
4.0
D_{mean} (connection strength)

Exp 3 — Effect of shape difference on winner-take-all (N=500)





exp3_dS_dEpsilon_vs_D.png



Comparison Experiments

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Cross-regime experiments and averaged heatmaps:

  Exp 1  — LC frequency competition (varying Δω)
  Exp 2a — FP rotation (varying eigenvector angle θ)
  Exp 2b — FP coupling (varying off-diagonal Δb)
  Exp 3  — Cross-comparison (LC-vs-LC, FP-vs-FP, Mixed)

  Averaged heatmaps of S(Δfeature, ΔD) for Δω and Δb
  S at ΔD = 0, sensitivity slopes at zero feature gap
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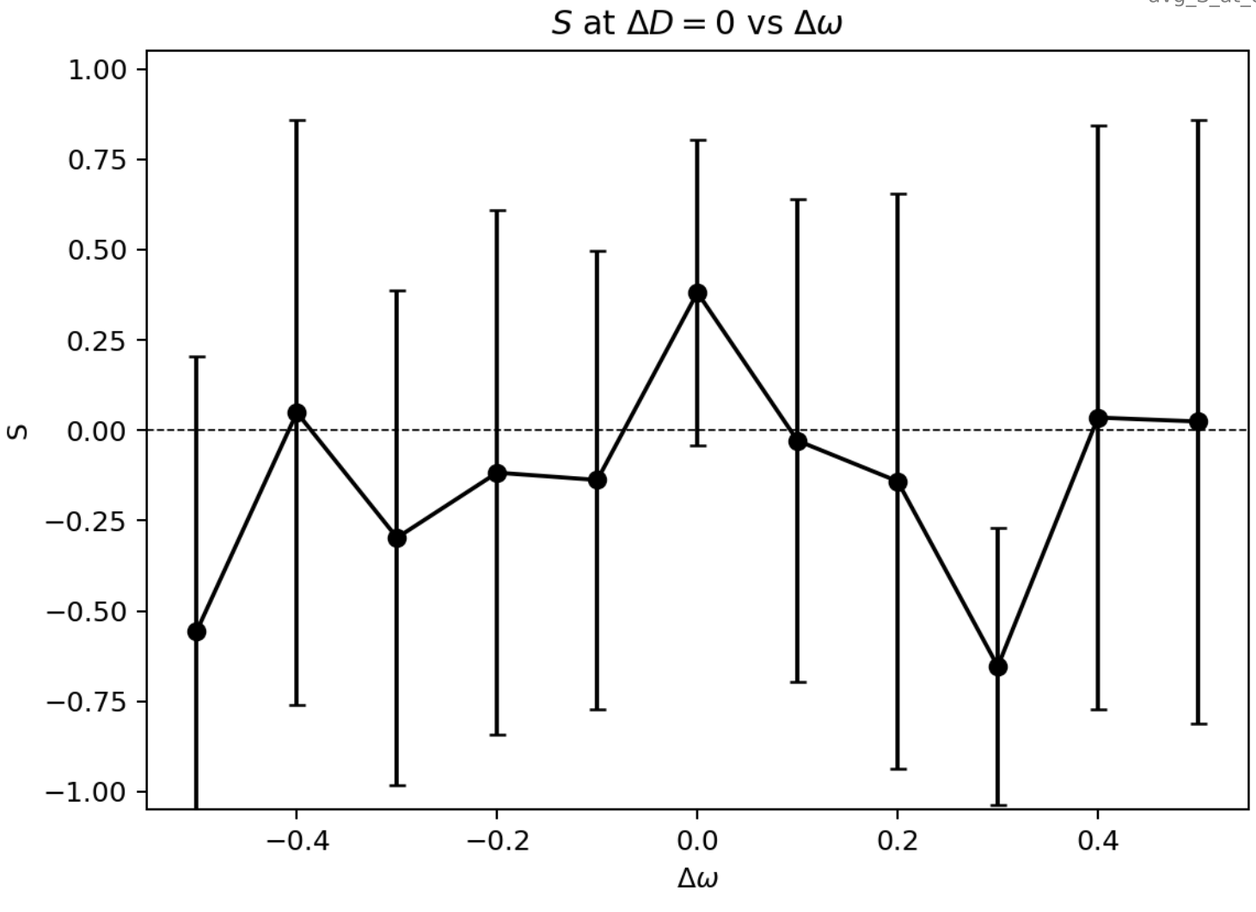
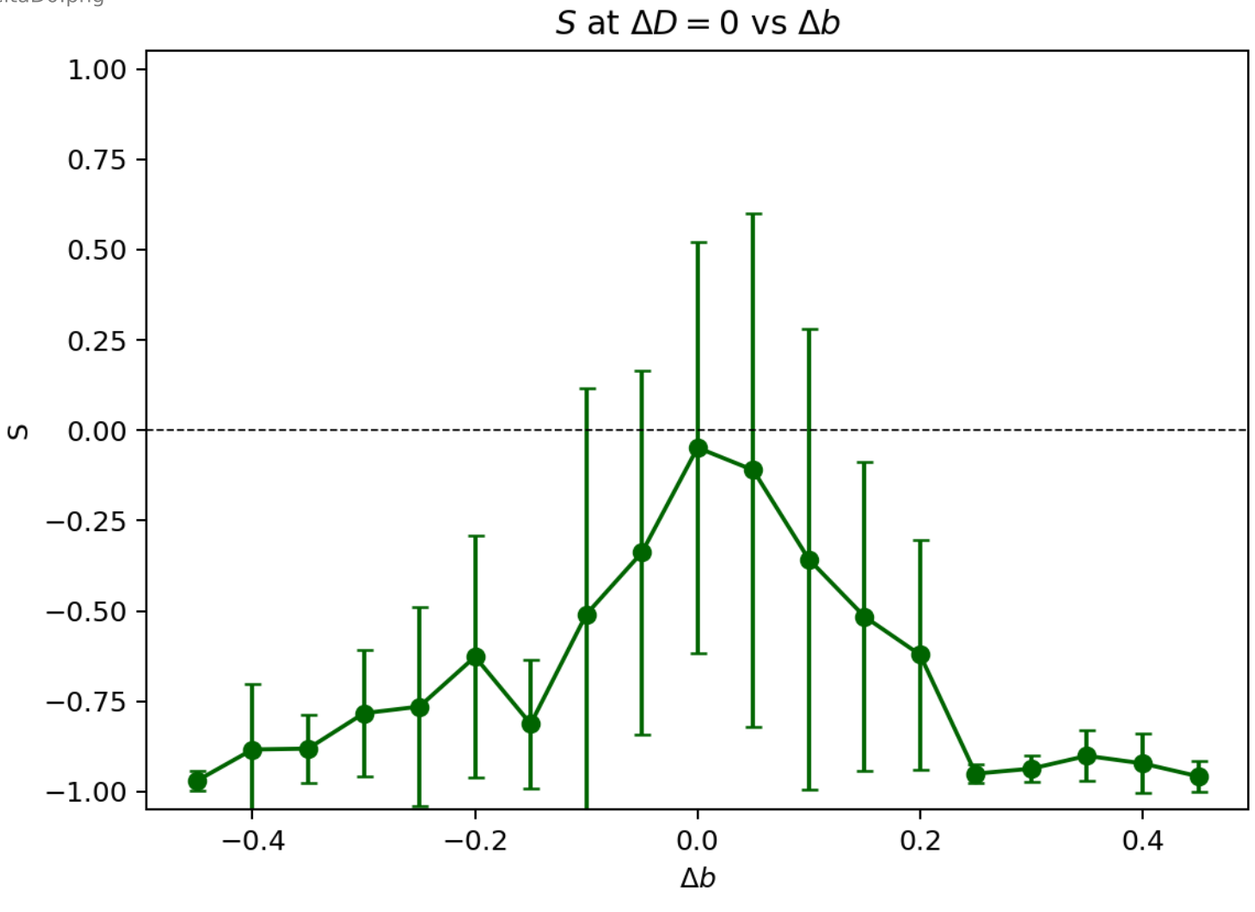
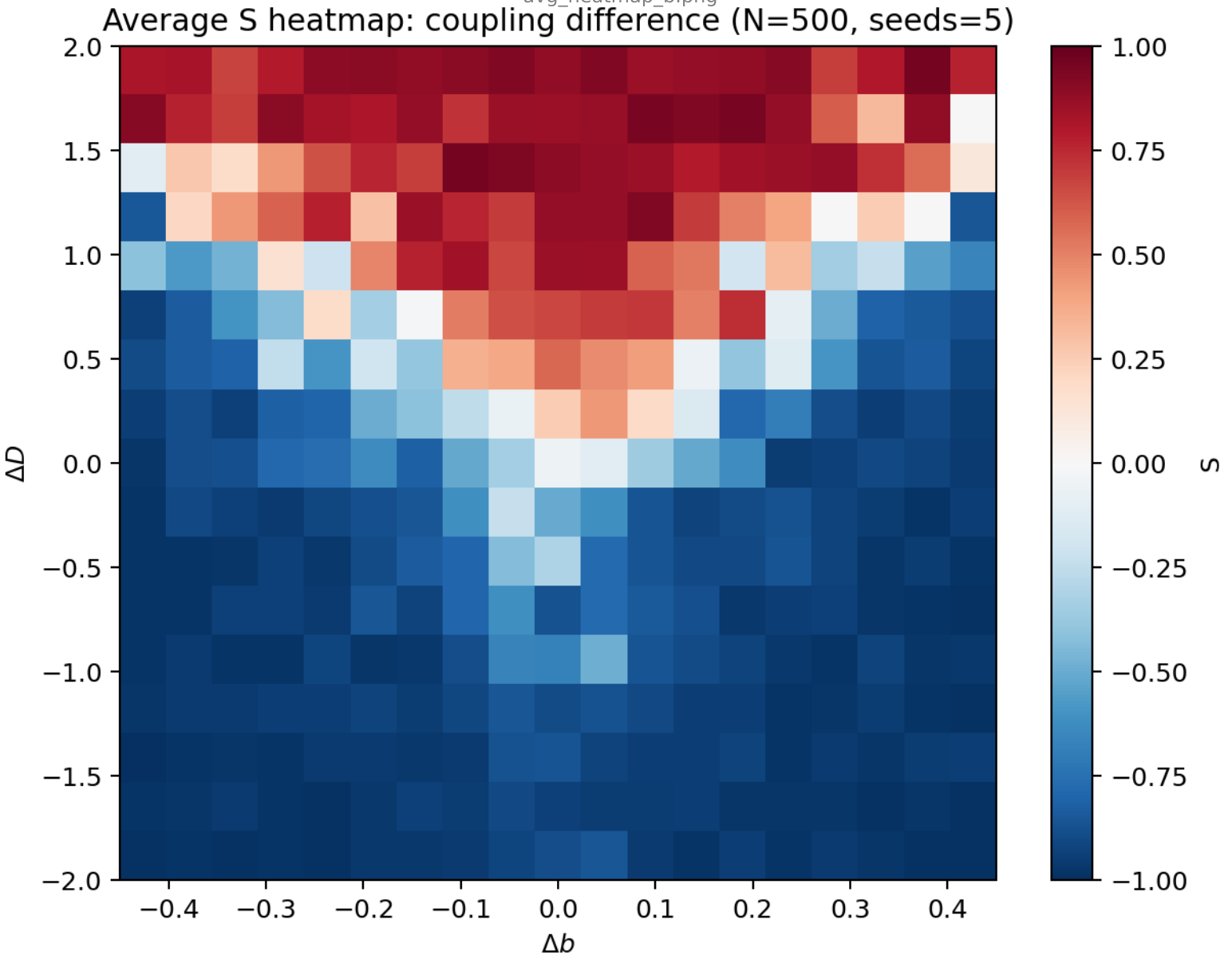
avg_S_at_deltaD0.png

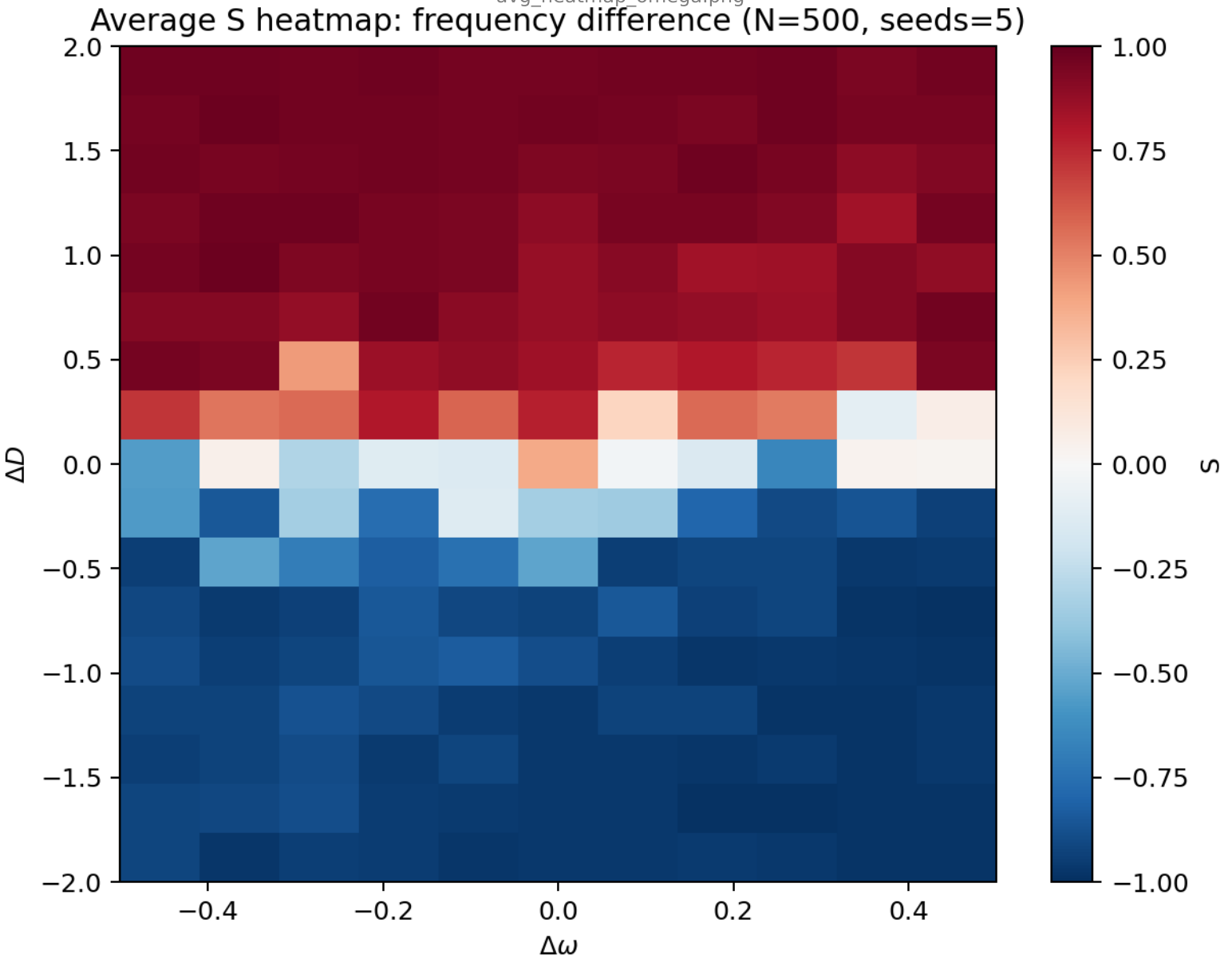




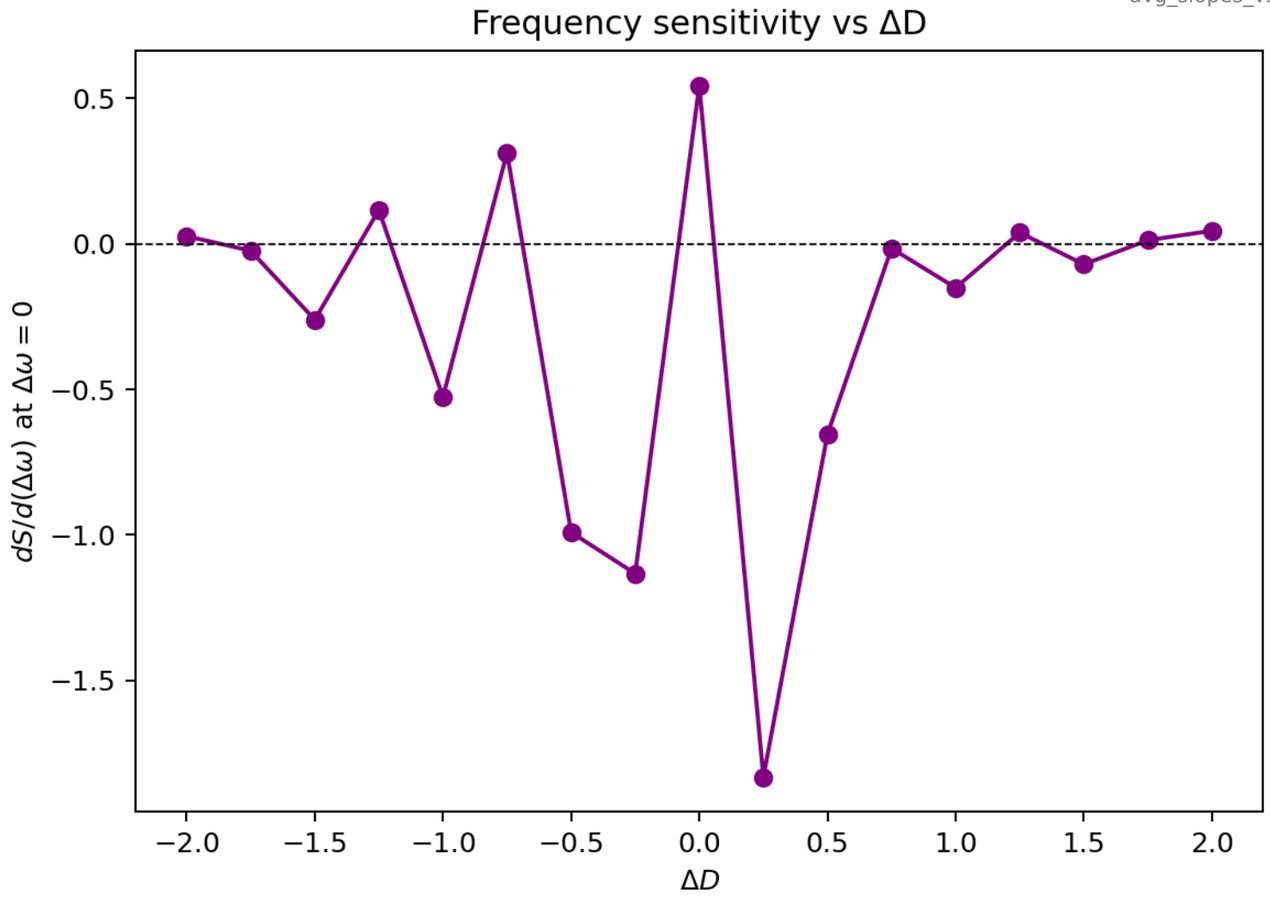
avg_heatmap_b.png

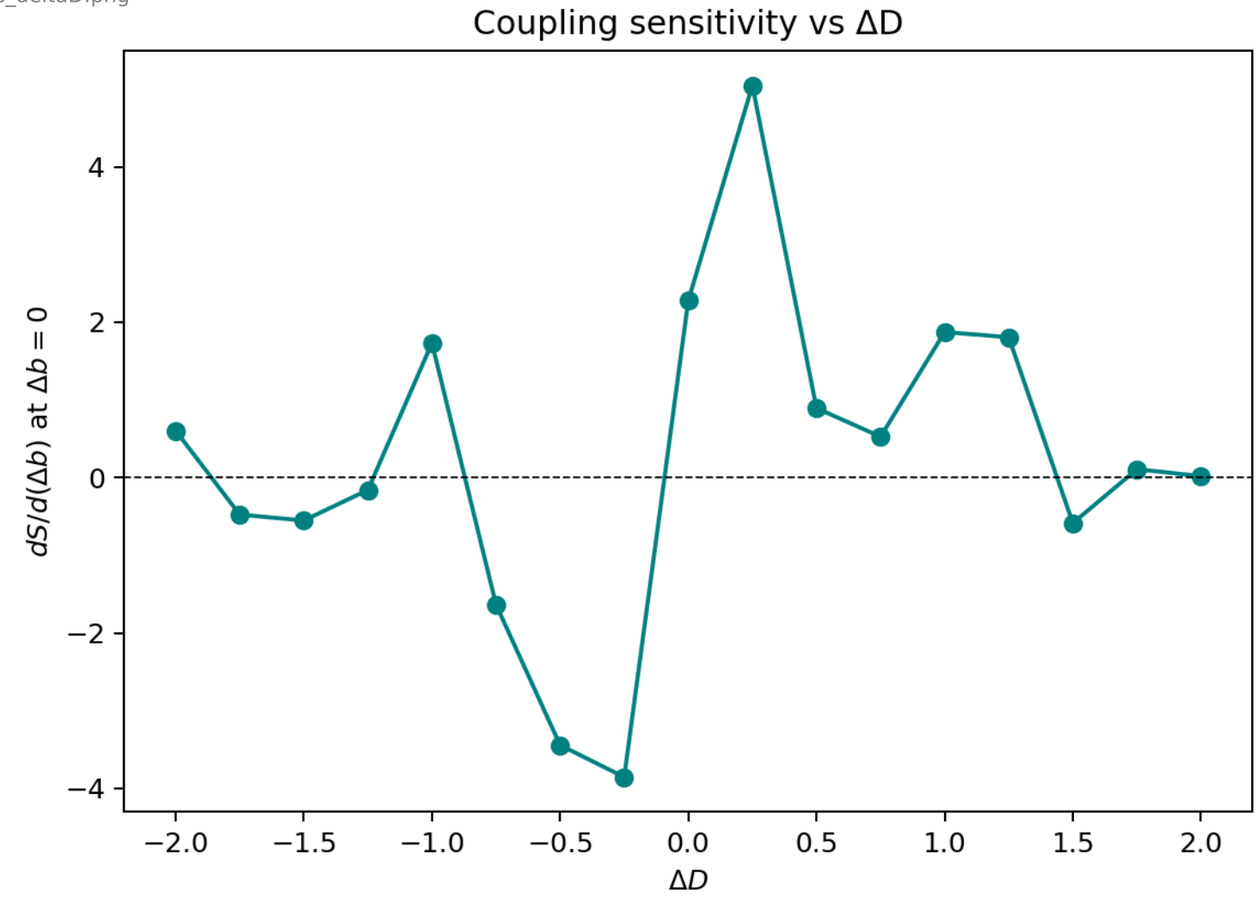


avg_heatmap_omega.png

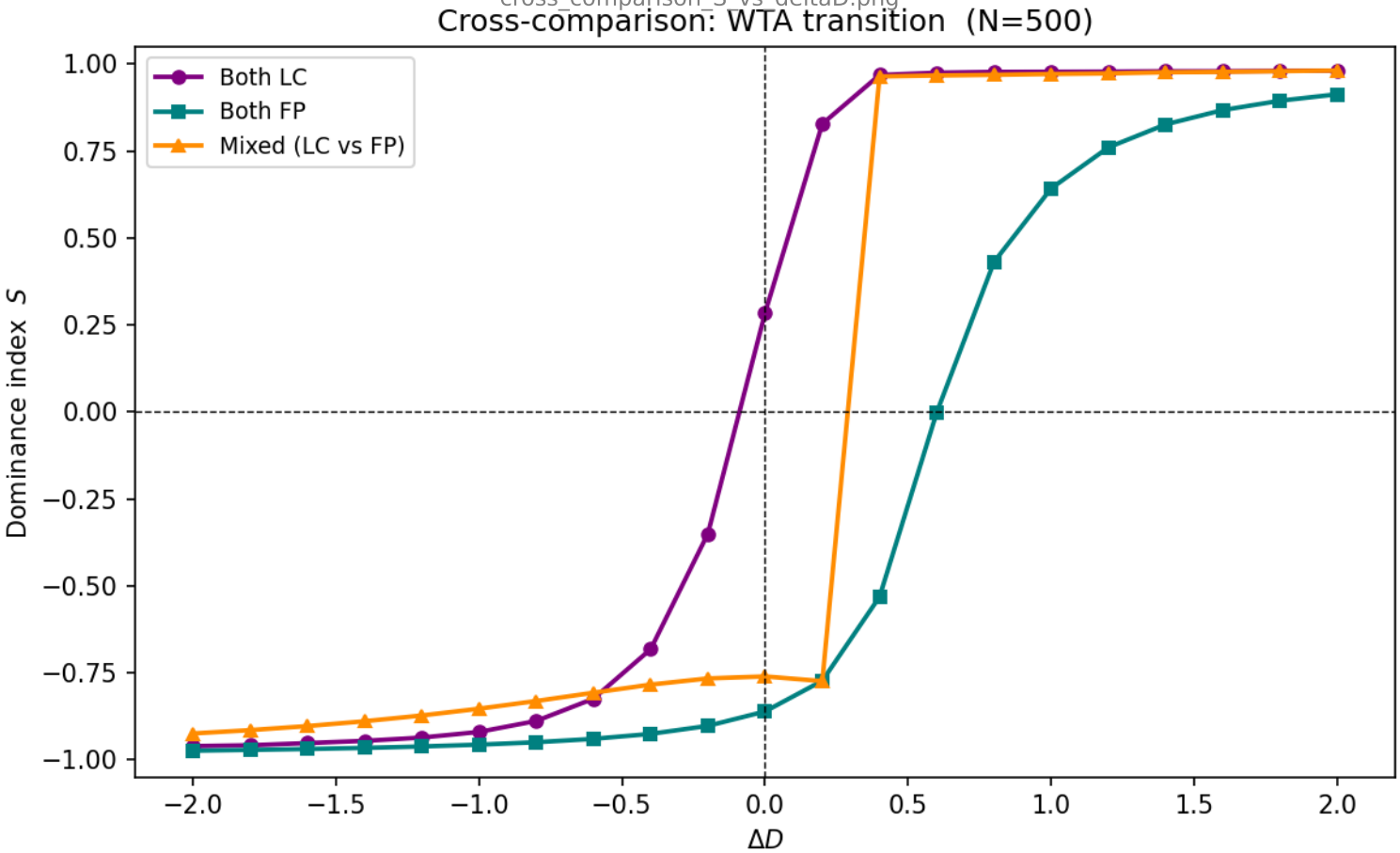


avg_slopes_vs_deltaD.png



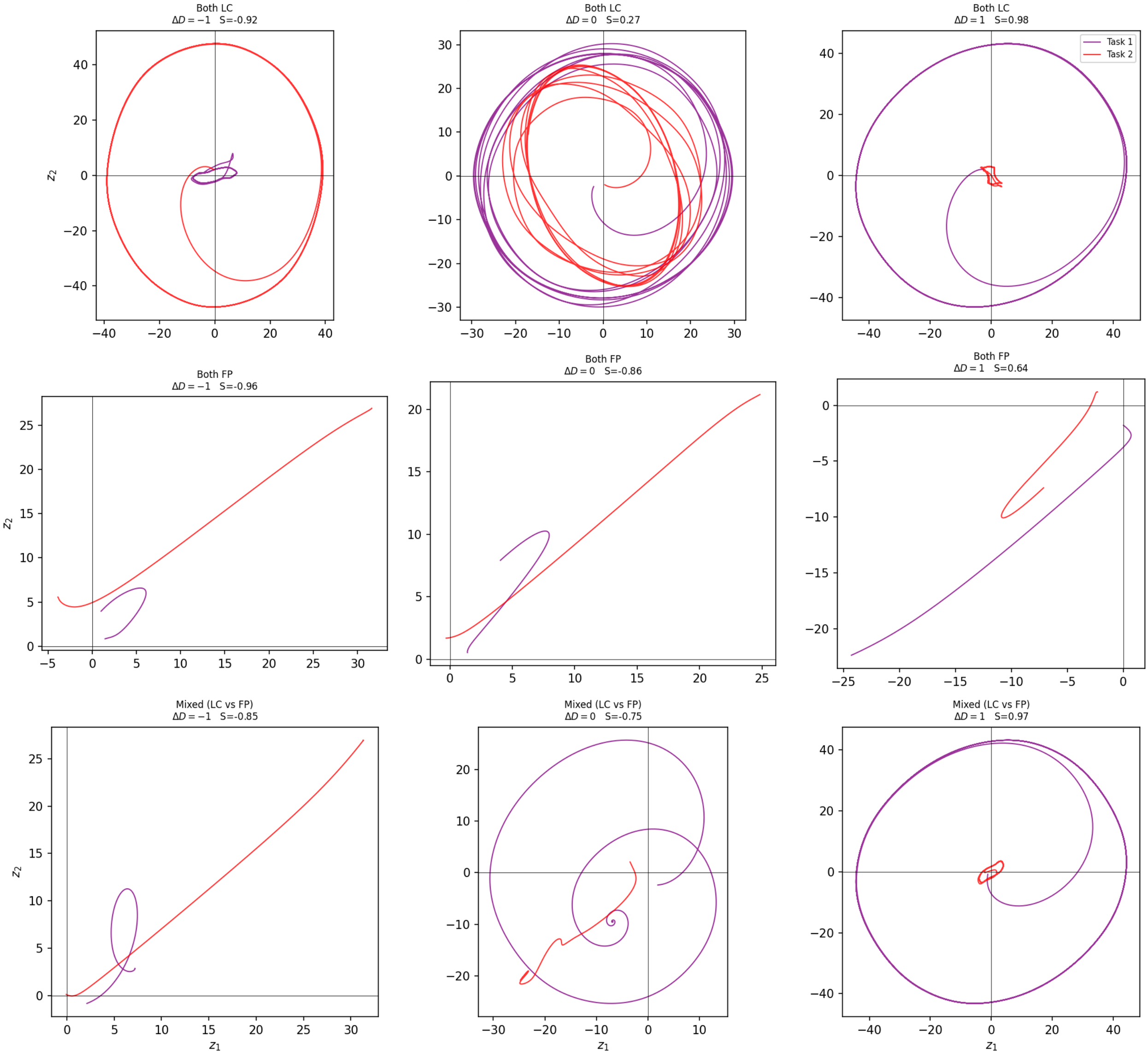


cross_comparison_S_vs_deltaD.png

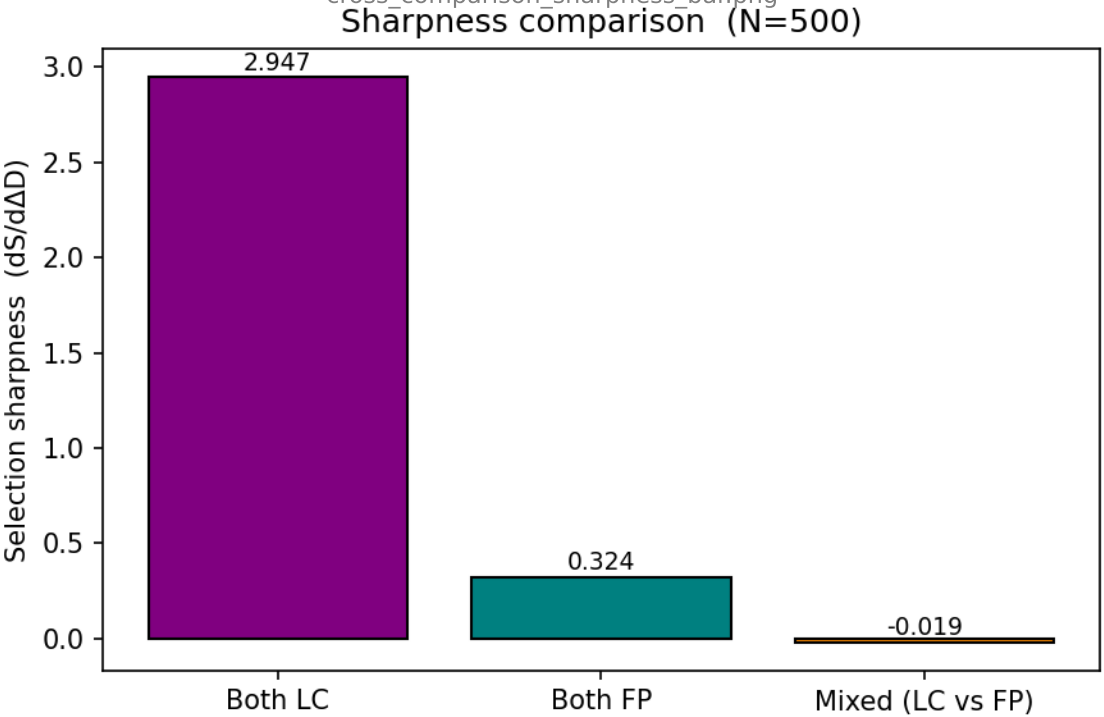


cross_comparison_phase_portraits.png

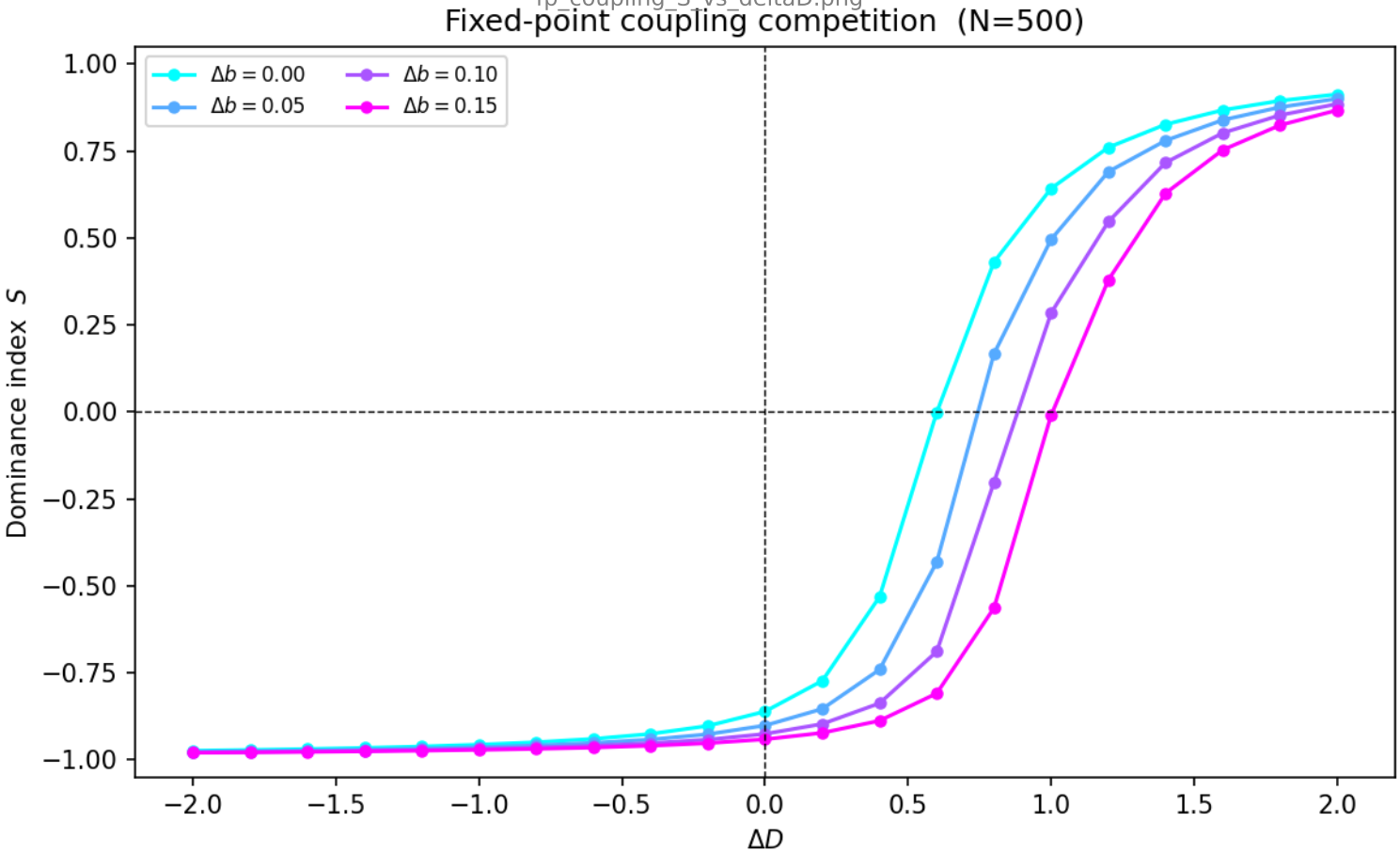
Phase portraits — Cross-comparison (N=500)



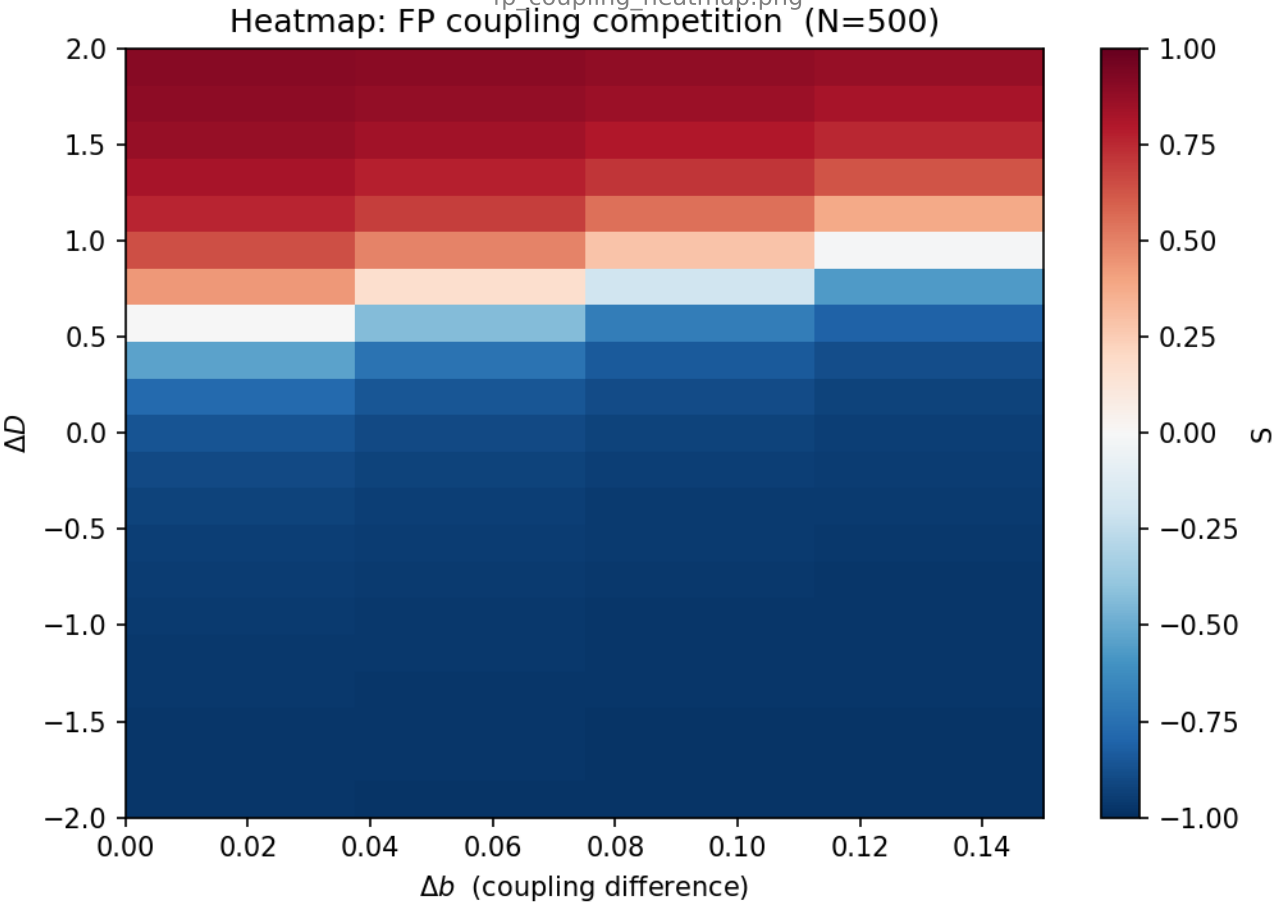
cross_comparison_sharpness_bar.png



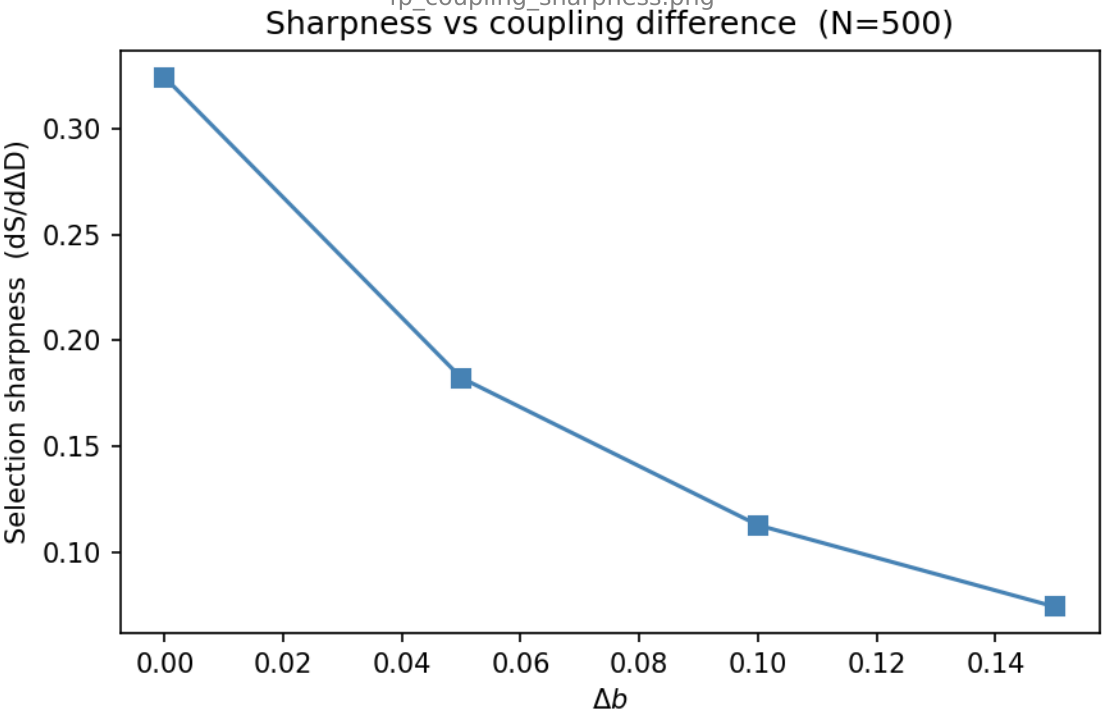
fp_coupling_S_vs_deltaD.png



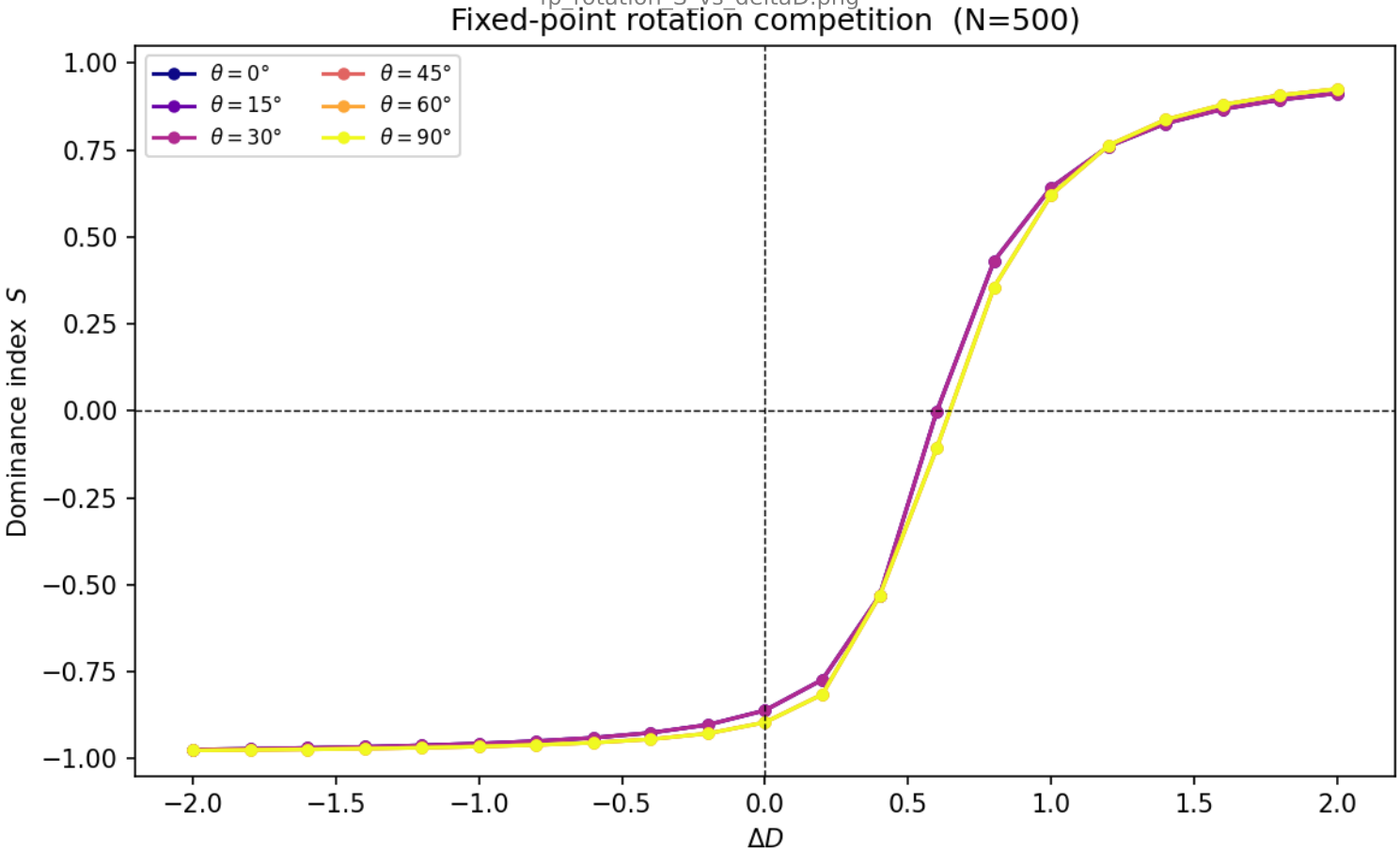
fp_coupling_heatmap.png

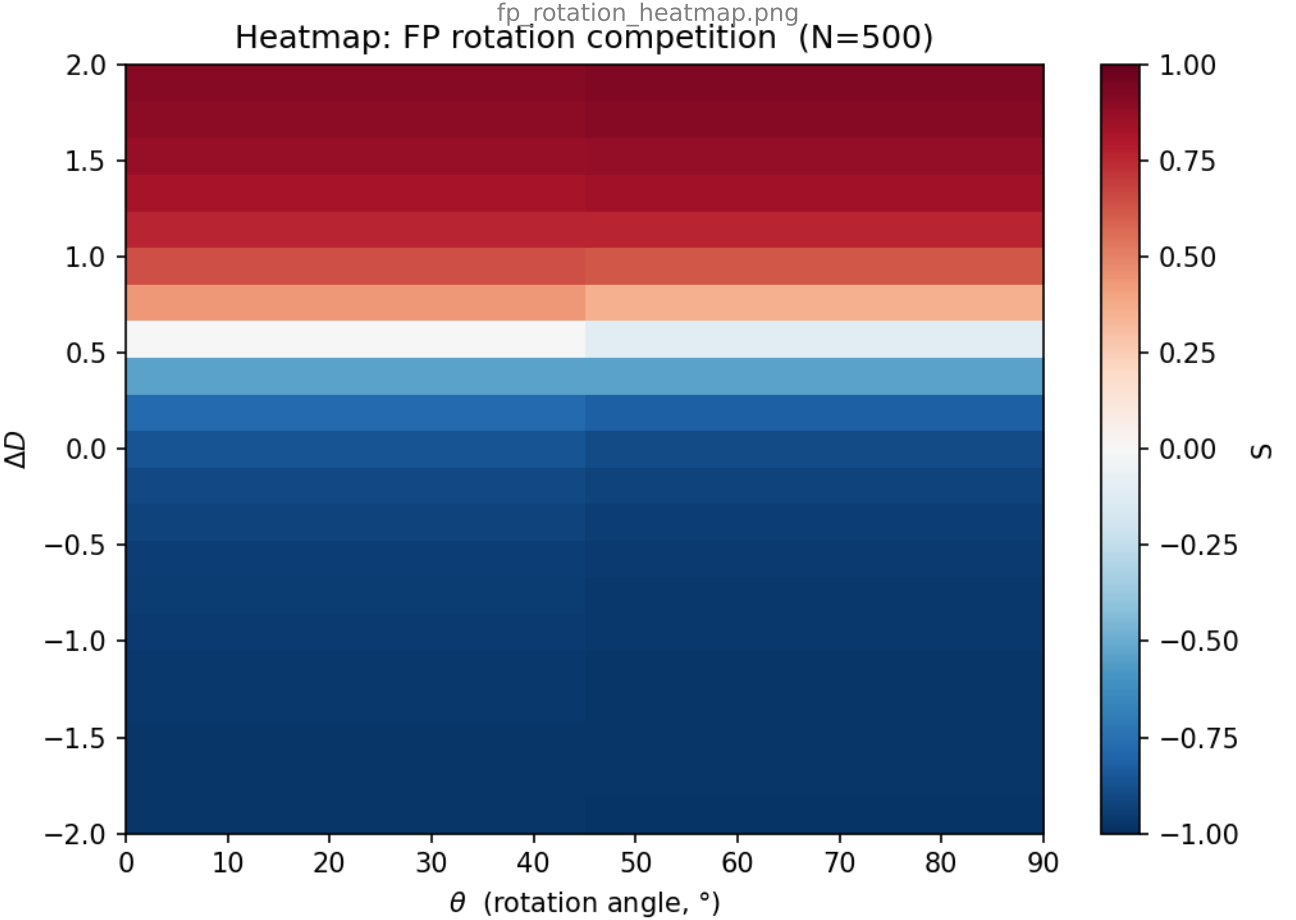


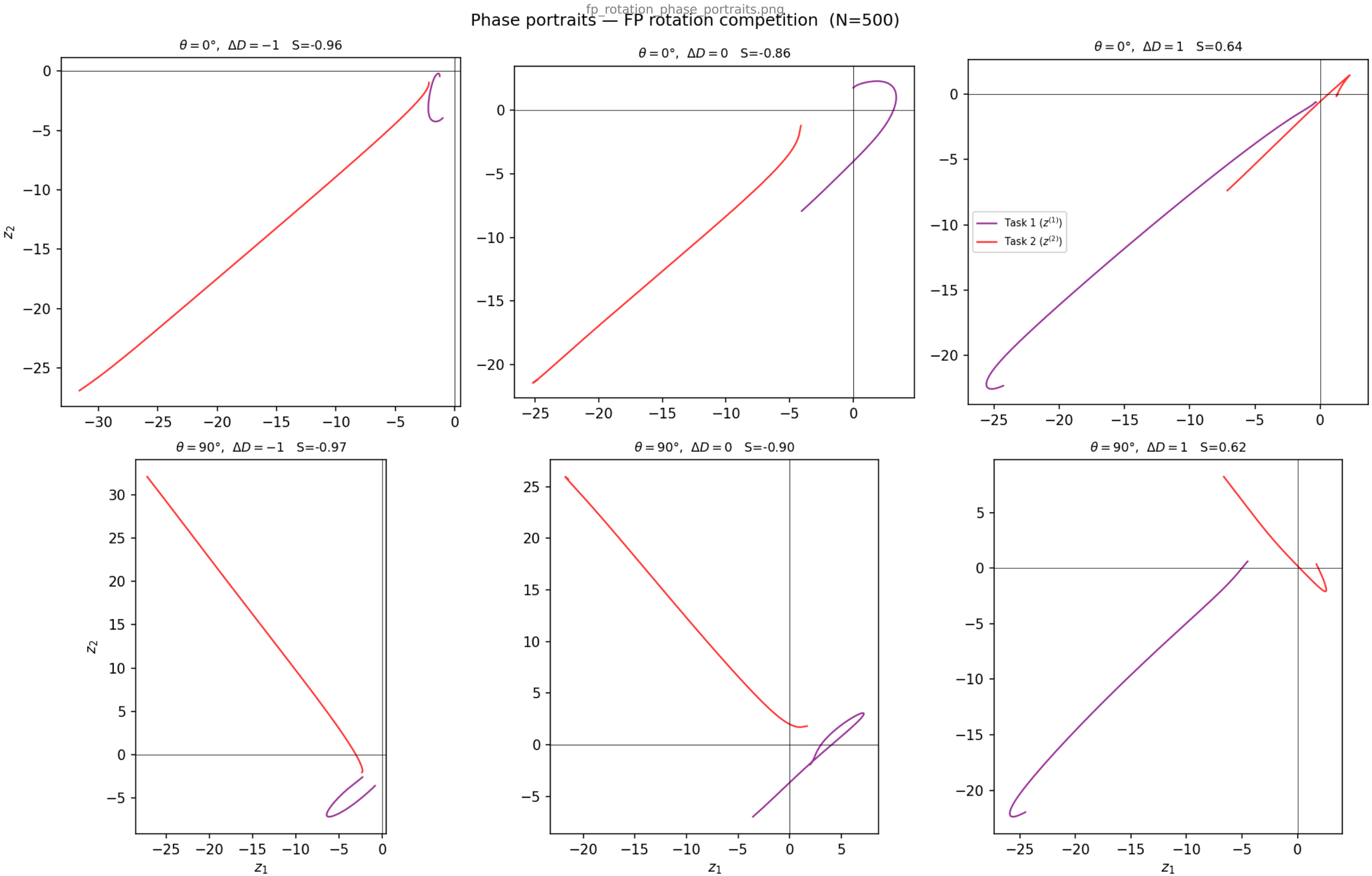
fp_coupling_sharpness.png



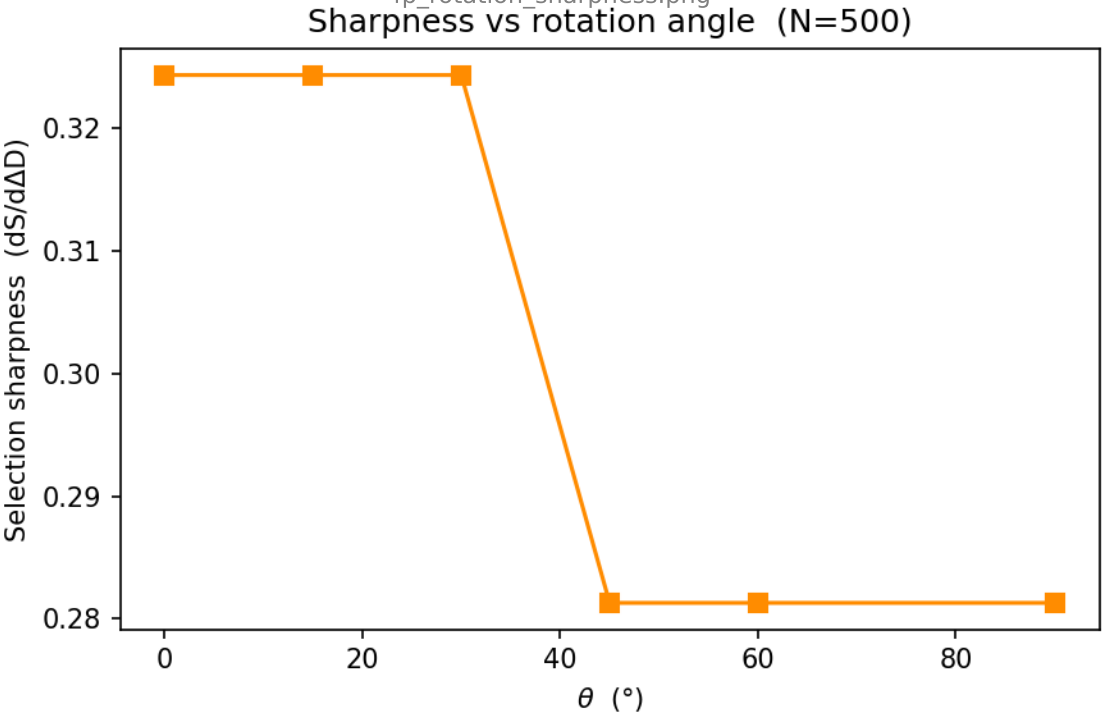
fp_rotation_S_vs_deltaD.png



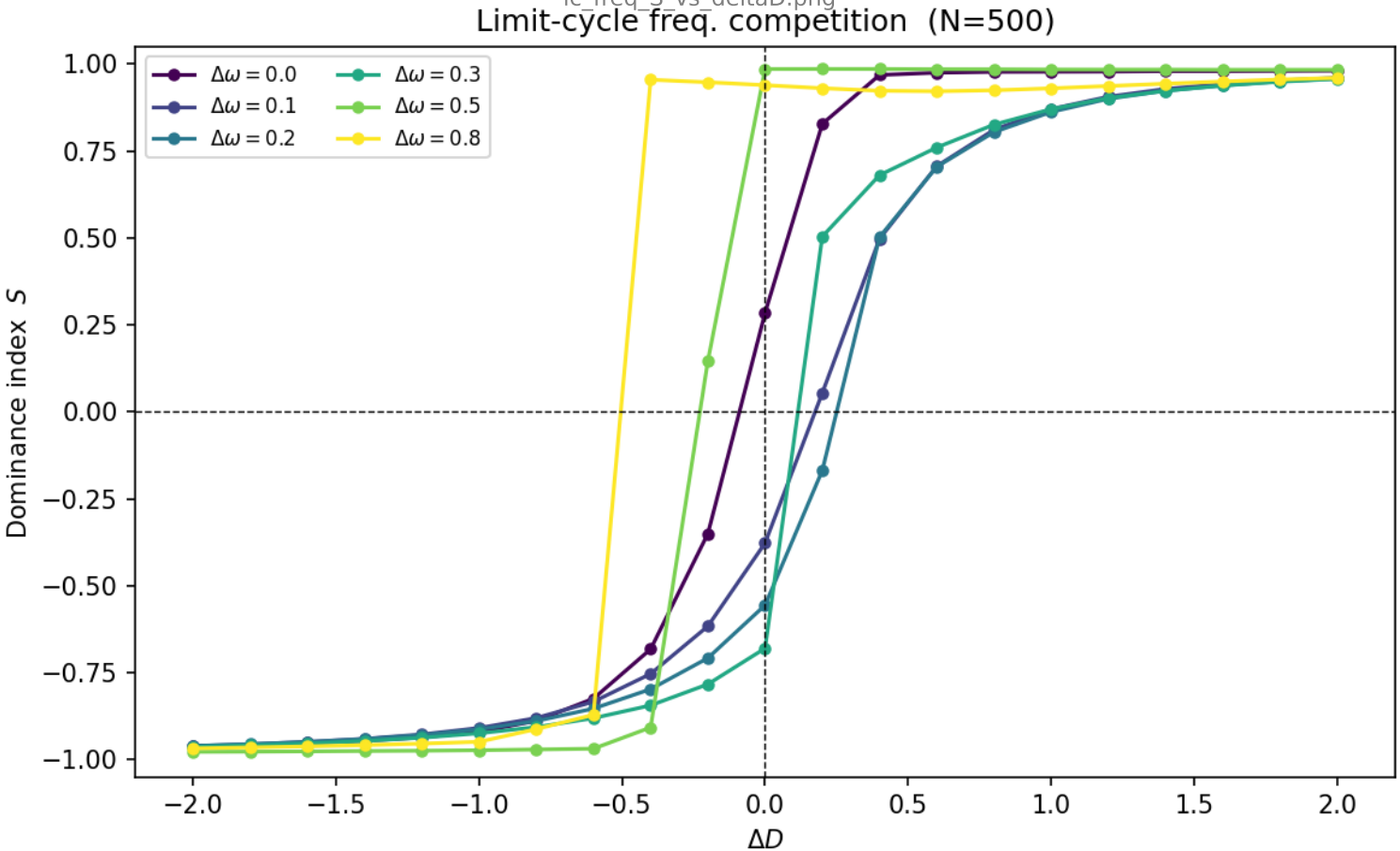

fp_rotation_heatmap.png
Heatmap: FP rotation competition (N=500)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
ΔD
0
10
20
30
40
50
60
70
80
90
θ (rotation angle, °)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
S


Phase portraits — FP rotation competition (N=500)
$\theta = 0°$, $\Delta D = -1$ S=-0.96
$\theta = 0°$, $\Delta D = 0$ S=-0.86
$\theta = 0°$, $\Delta D = 1$ S=0.64
$\theta = 90°$, $\Delta D = -1$ S=-0.97
$\theta = 90°$, $\Delta D = 0$ S=-0.90
$\theta = 90°$, $\Delta D = 1$ S=0.62
Task 1 ($z^{(1)}$)
Task 2 ($z^{(2)}$)
z_1
z_2

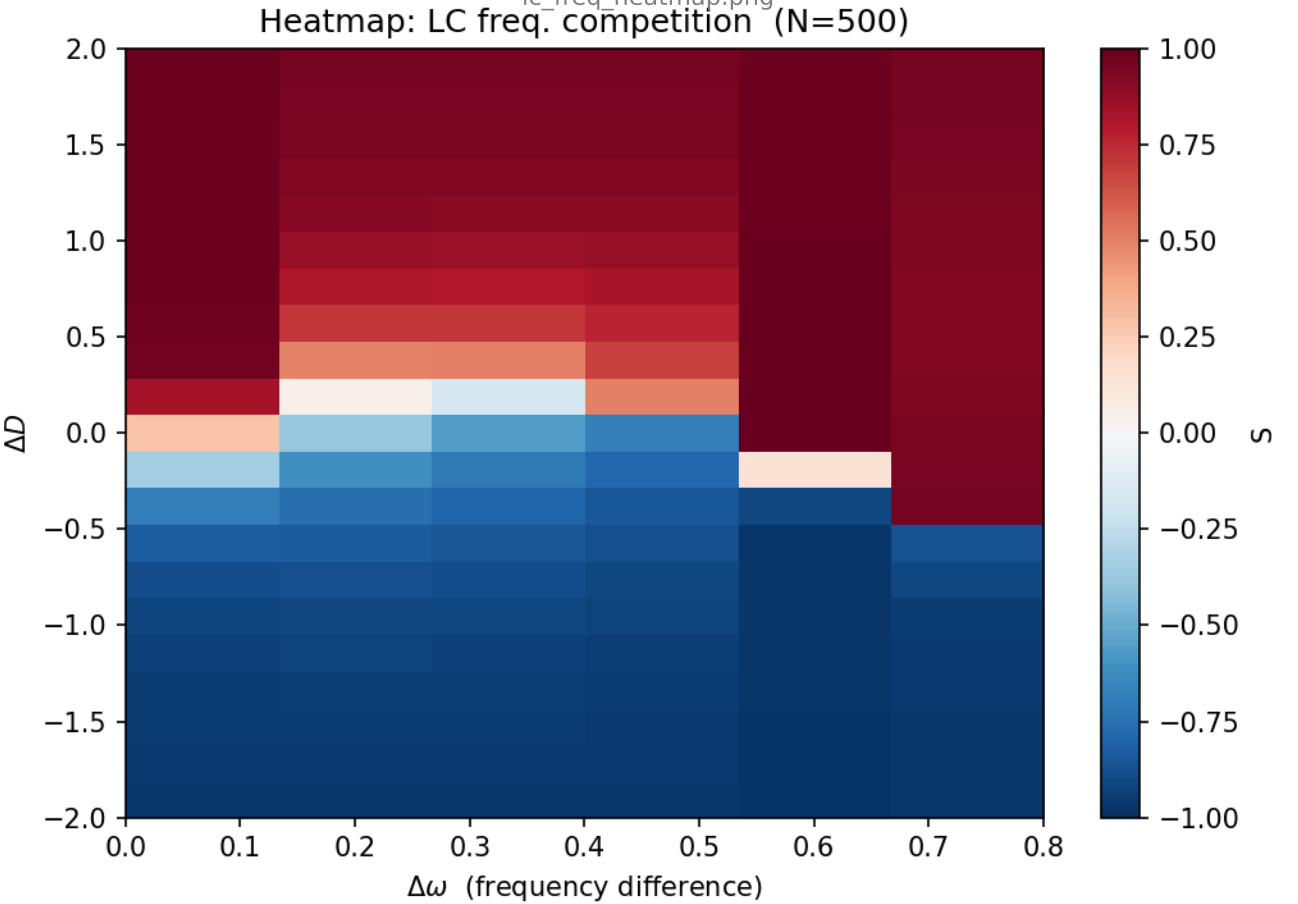
fp_rotation_sharpness.png



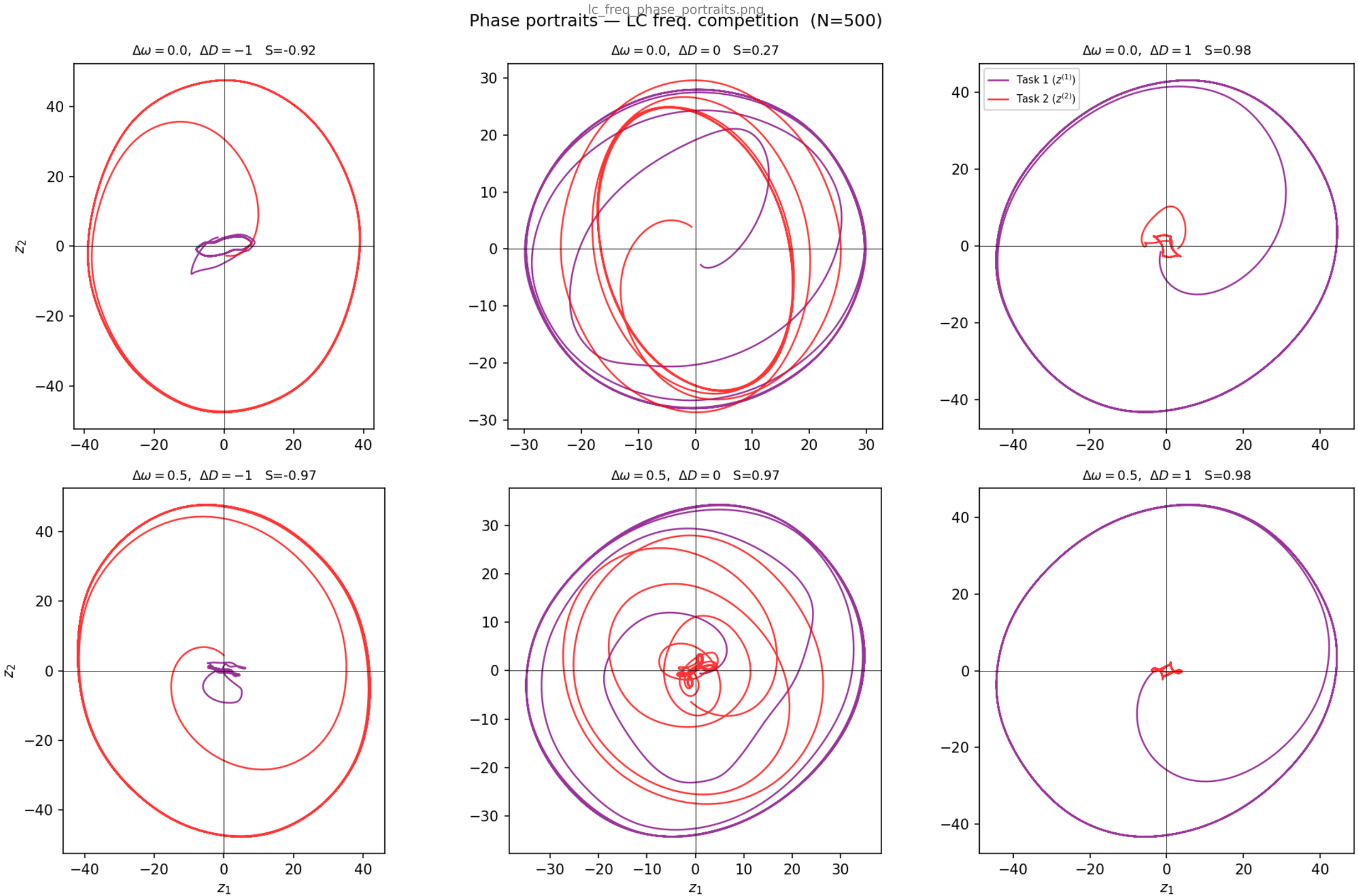
lc_freq_S_vs_deltaD.png



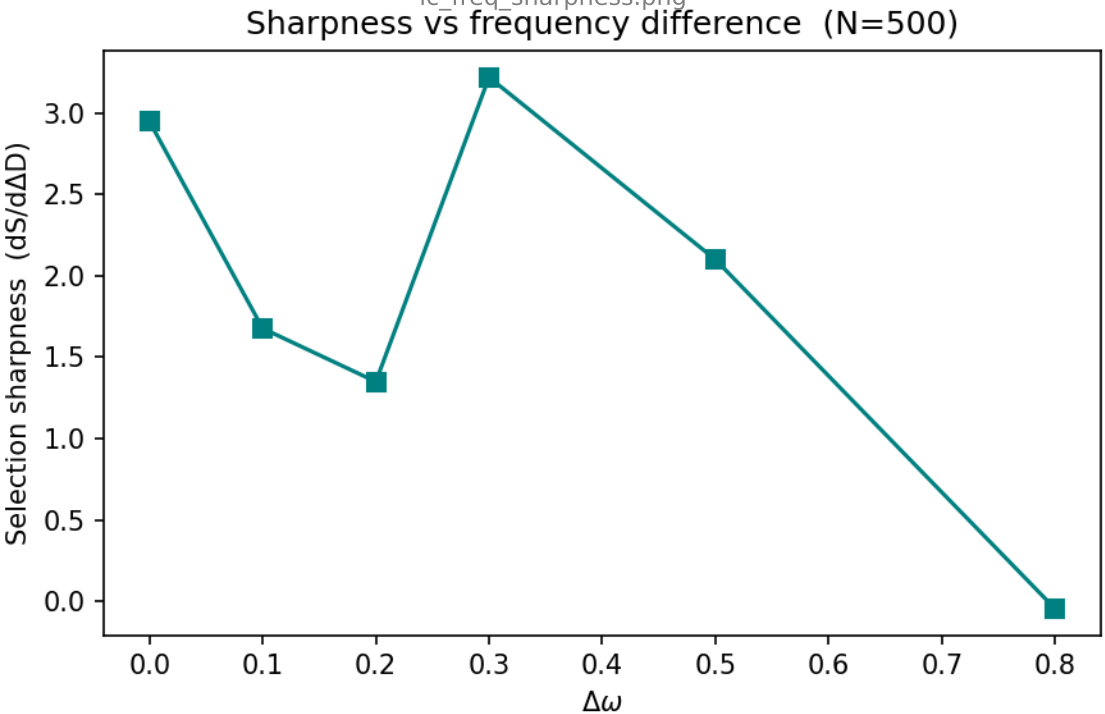
lc_freq_heatmap.png

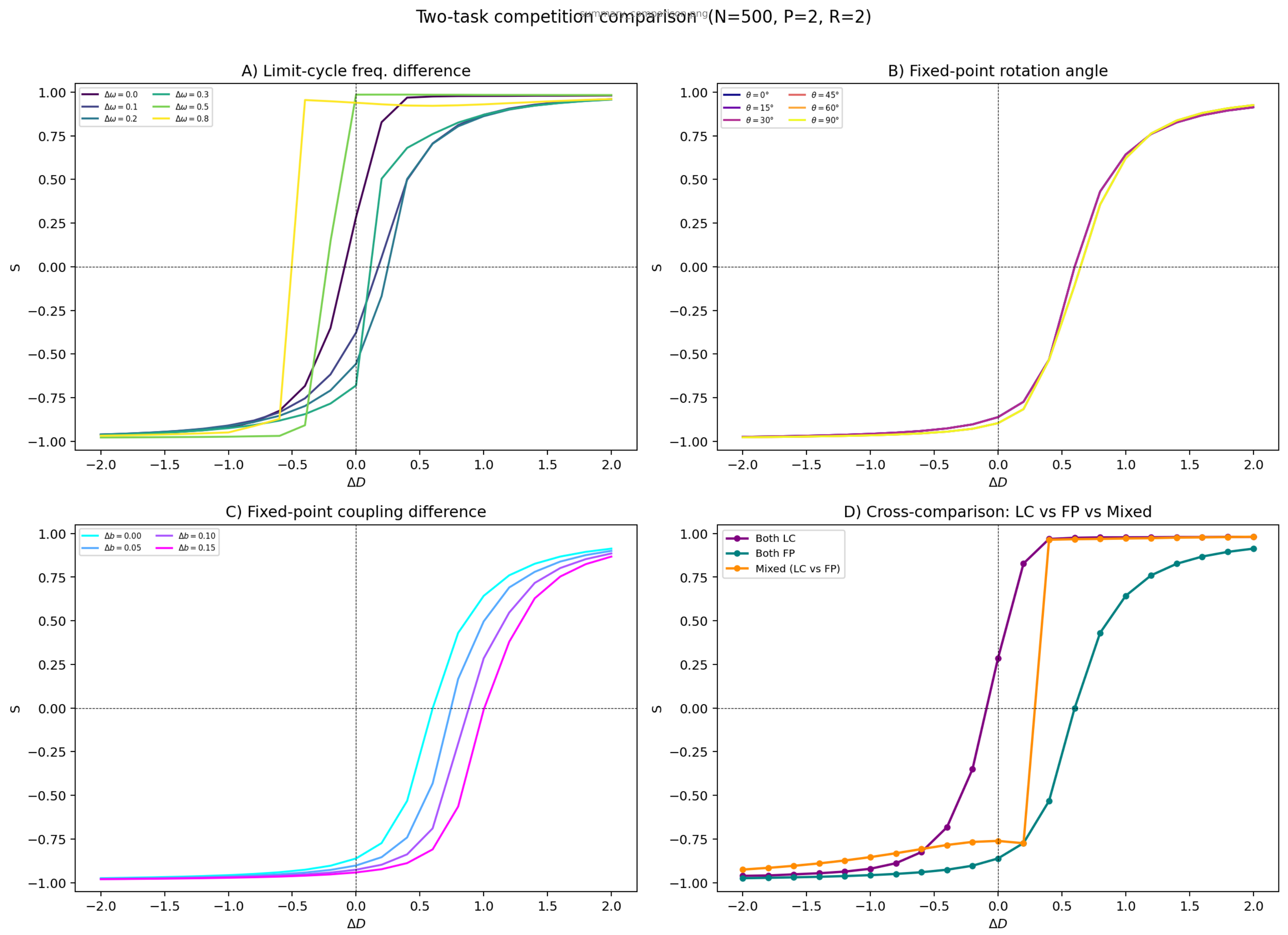


lc_freq_phase_portraits.png



lc_freq_sharpness.png




Two-task competition comparison (N=500, P=2, R=2)
summary_comparison.png
A) Limit-cycle freq. difference
Δω = 0.0
Δω = 0.1
Δω = 0.2
Δω = 0.3
Δω = 0.5
Δω = 0.8
B) Fixed-point rotation angle
θ = 0°
θ = 15°
θ = 30°
θ = 45°
θ = 60°
θ = 90°
C) Fixed-point coupling difference
Δb = 0.00
Δb = 0.05
Δb = 0.10
Δb = 0.15
D) Cross-comparison: LC vs FP vs Mixed
Both LC
Both FP
Mixed (LC vs FP)
S
ΔD